

Queue

Q.N0.1

Problem Description

The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A , which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i .

Note: Two numbers can be the same value-wise but they should be distinct index-wise.

Constraints:

$1 \leq N \leq 100000$
 $0 \leq A[i] \leq 1000000$

Input:

The first line contains an integer N , denoting the number of elements in the array A .
The next line contains N space separated integers, each denoting the i th integer of the array A .

Output:

Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.

Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2 3 4 5

EXPECTED OUTPUT

-1
-1
6
24
60

Test Case 2

INPUT (STDIN)

8
3 7 1 2 4 7 5 9

EXPECTED OUTPUT

-1
-1
21
42
84
196
245
441

Mandatory Test Cases

Test Case 1

KEYWORD

if(biggest<big)

Test Case 2

KEYWORD

if(a[i]>biggest)

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n;
```

```
int a[100000];

printf("Input value:\n");

scanf("%d", &n);

for(int i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}

printf("\nOutput:\n");

for(int i = 0; i < n; i++)
{
    int biggest = -1;
    int big = -1;
    int third = -1;
    for(int j = 0; j <= i; j++)
    {
        if(a[j] > biggest)
        {
            third = big;
            big = biggest;
            biggest = a[j];
        }
        else if(a[j] > big)
        {
            third = big;
            big = a[j];
        }
        else if(a[j] > third)
        {
            third = a[j];
        }
    }

    if(biggest < big)
    {
        printf("-1\n");
    }
    else if(i < 2)
    {
        printf("-1\n");
    }
    else
    {
        long long product = (long long)biggest * big * third;
        printf("%lld\n", product);
    }
}

return 0;
```

```
}
```

Output;

Test_case 1:

```
Input value:
```

```
5
```

```
1 2 3 4 5
```

```
Output:
```

```
-1
```

```
-1
```

```
6
```

```
24
```

```
60
```

Test_case 2:

```
Input value:
```

```
8
```

```
3 7 1 2 4 7 5 9
```

```
Output:
```

```
-1
```

```
-1
```

```
21
```

```
42
```

```
84
```

```
196
```

```
245
```

```
441
```